

Session: B

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Effect of Force-Rate on Continuous Kinesthetic Force Discrimination

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Introduction

- Conventional haptic force discrimination tasks involve perceptual differentiation between stimulus pairs separated by a brief inter-stimulus interval (ISI).
- It is unclear whether the **Force JND** measured using conventional methods, i.e., with ISI in between stimuli, remains consistent in the case of **Continuous Force Discrimination**.
- Weber Fraction, $\delta = \frac{|F_t - F_r|}{|F_r|}$, where, F_r = Reference Force, F_t = test force

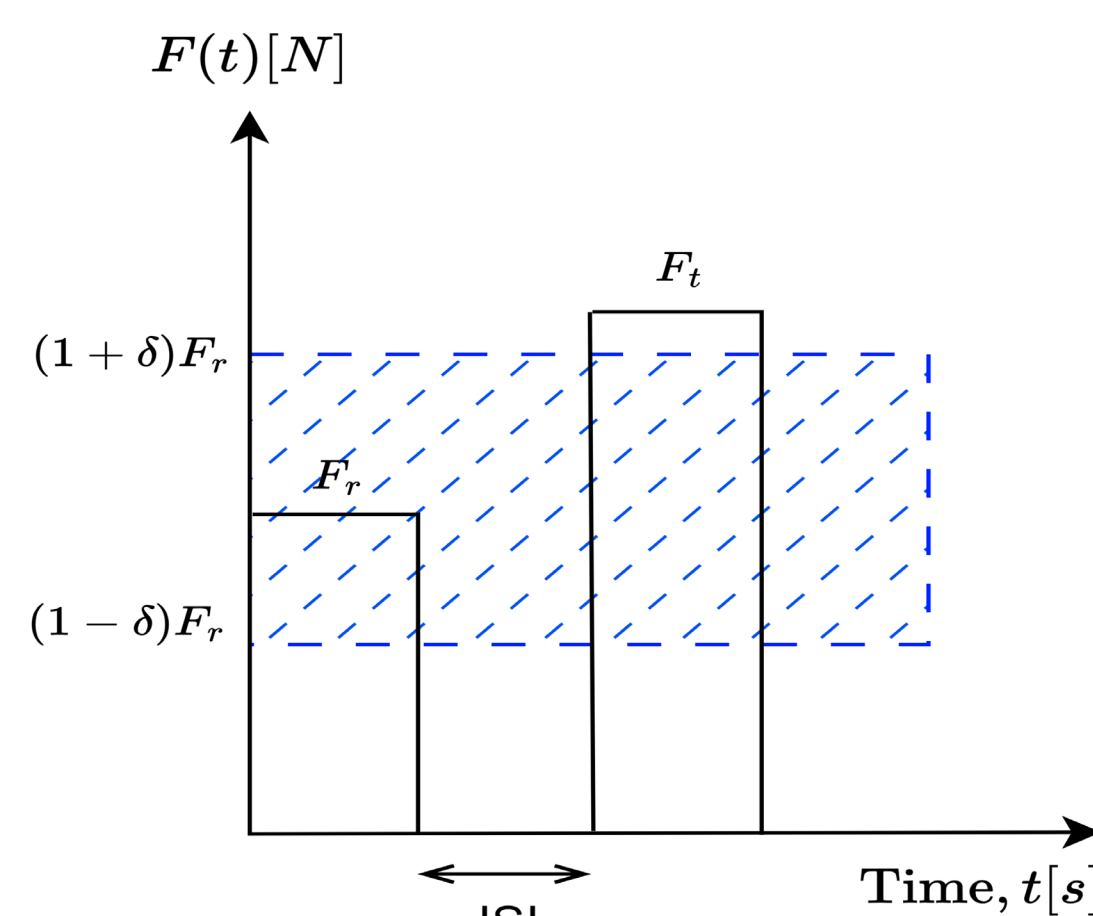
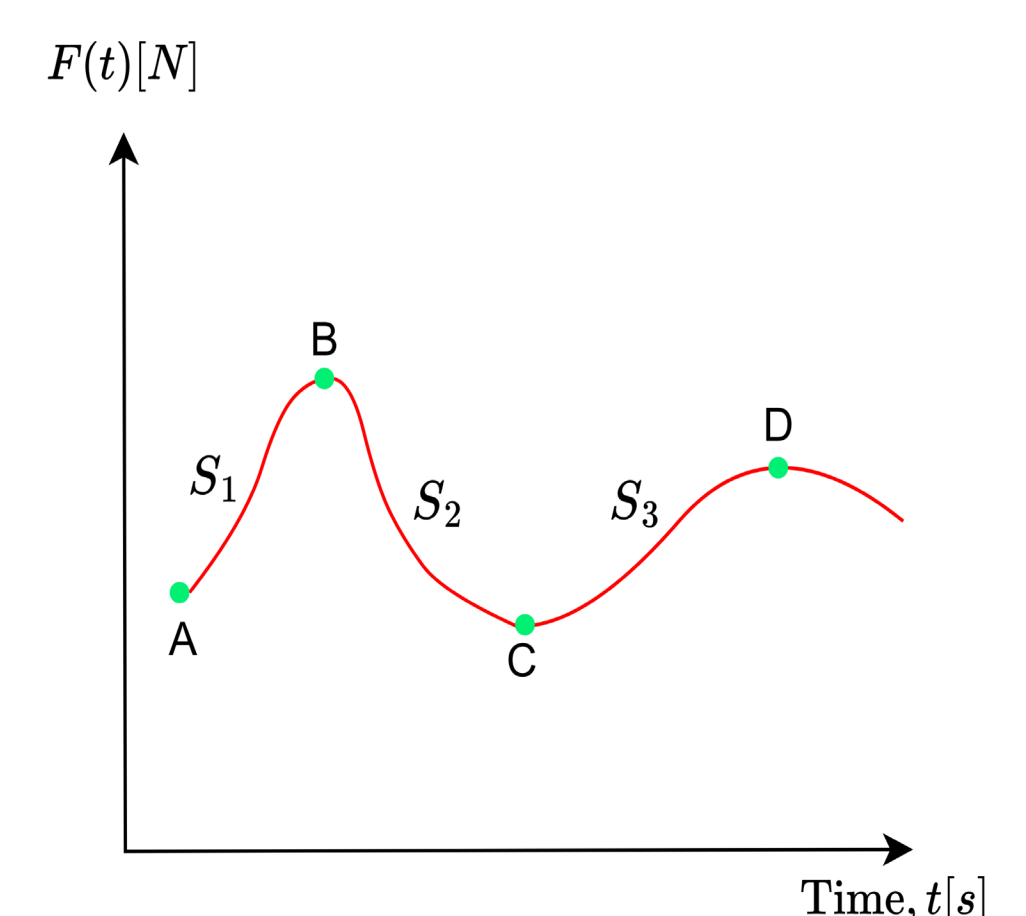


Fig.1: Force discrimination with ISI and corresponding deadzone (dashed area)

Fig.2: Continuous force stimuli $F(t)$ with varying force-rates S_1 , S_2 , and S_3

Methodology

- Machine learning classification-based approach
- Paradigm: Participant is exposed to a linearly increasing 1D kinesthetic force stimulus and is asked to react to the change in force.

❖ Setup

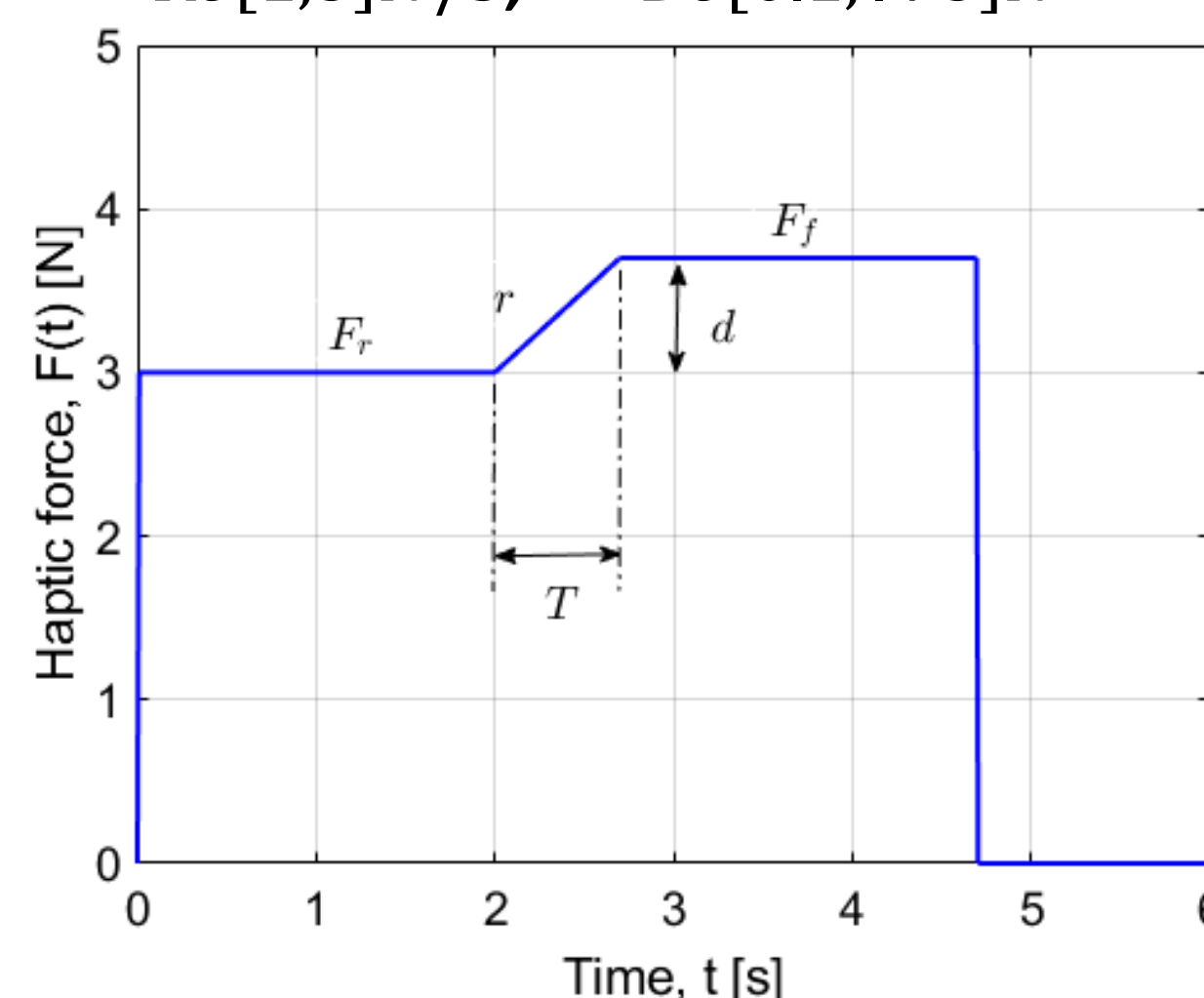
- PHANToM Premium 1.5 HF haptic device
- One-dimensional (1D) haptic force stimulus $\rightarrow +Z$ -axis



Fig.3: Experimental setup

❖ Stimuli

- $R \in [1, 5] \text{ N/s}$, $D \in [0.1, .75] \text{ N}$

Fig.4: Force stimuli: Transition with force-rate (r)

❖ Data-Collection

- Participants: 10 users
- Total trials: 1000 trials per user

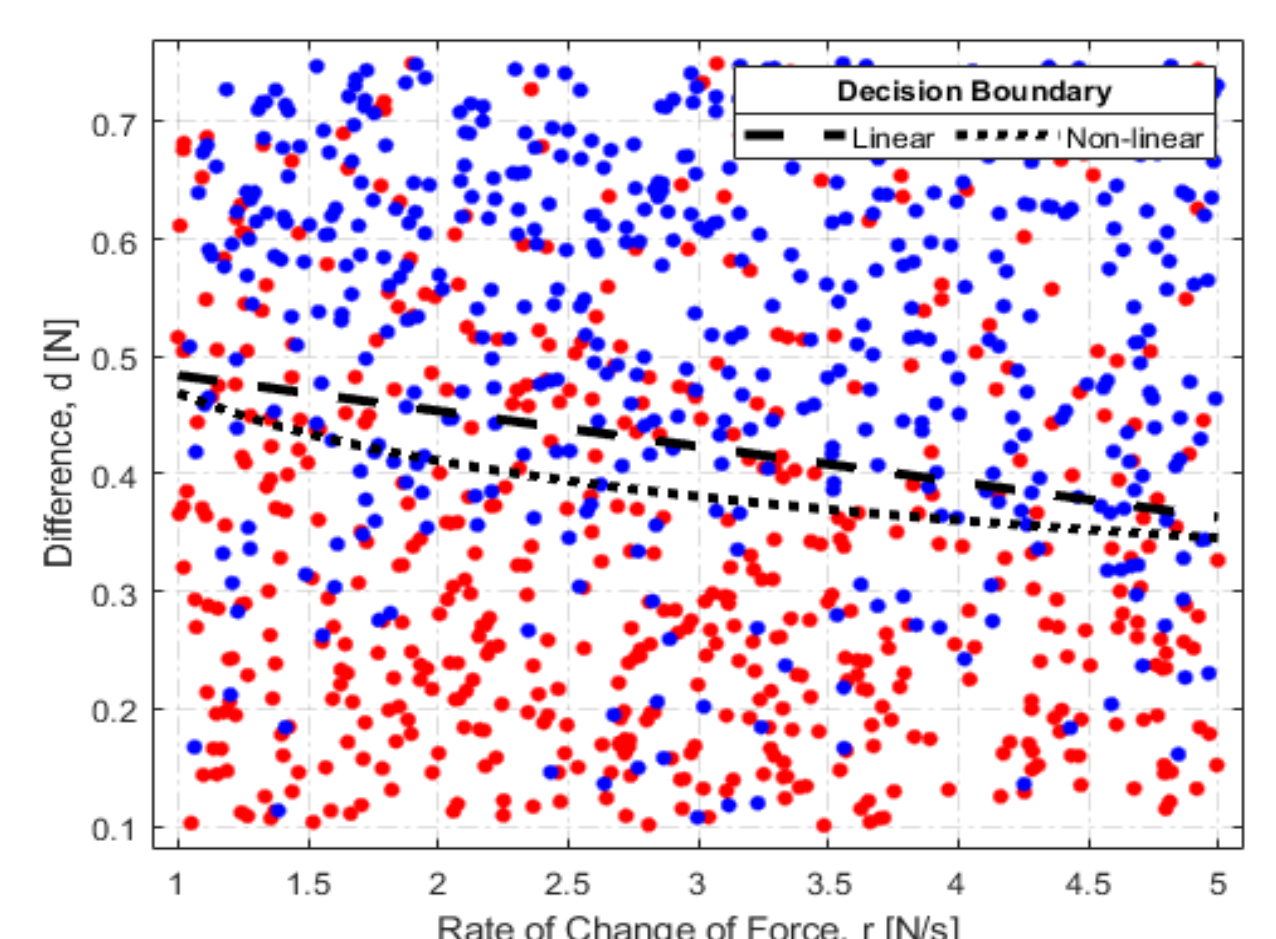


Fig. 5: Scatter plot for one participant

Estimation of Decision Boundary

$$E^k = \frac{1}{N} \sum_{i=1}^N \left(w_p \frac{(1+y_i)}{2} + w_{np} \frac{(1-y_i)}{2} \right) \mathbf{1}(y_i \neq h^k(r_i, d_i))$$

$$w_p = \frac{N_{np}}{N_p + N_{np}} \quad \text{and} \quad w_{np} = \frac{N_p}{N_p + N_{np}}$$

1. Linear function-based boundary

$$d = mr + c$$

$$h^l(r_i, d_i) = \begin{cases} 1 & \text{for } d_i - (mr_i + c) \geq 0 \\ -1 & \text{for } d_i - (mr_i + c) < 0. \end{cases}$$

2. Power function-based boundary

$$d = \beta r^\alpha$$

$$h^p(r_i, d_i) = \begin{cases} 1 & \text{for } (d_i - \beta r_i^\alpha) \geq 0 \\ -1 & \text{for } (d_i - \beta r_i^\alpha) < 0 \end{cases}$$

Results and Analysis

- Boundary structures are validated using a Random Forest Classifier
- Mean prediction error $E^l = 27.94\%$, $E^p = 27.11\%$
- No significant difference in error ($t=1.388$, $p=0.198$)
- Two-way ANOVA with factors: *force-rate* and *perceptual boundaries*
 - A significant effect of force-rate on Weber fraction ($F(2,54) = 55.27$, $p < 0.001$)
 - A significant effect of perceptual boundaries on Weber fraction ($F(1,54) = 33.66$, $p < 0.001$)
 - The interaction effect between force-rate and perceptual boundaries is not significant ($F(2,54) = 0.83$, $p = 0.440$)

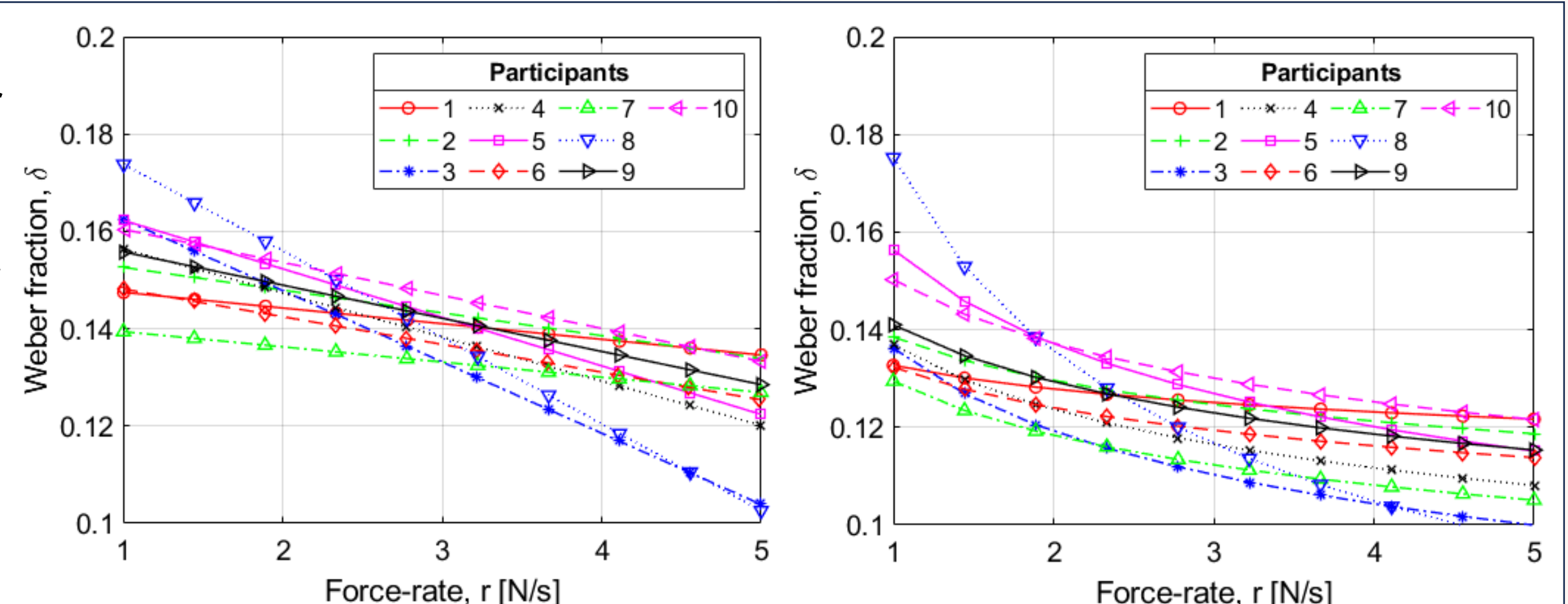


Fig.6: (a) Linear, (b) Non-linear boundaries

Conclusion

- The force-rate (r) significantly affects haptic force perception and δ decreases with the positive rate of change of force in the range of $[1, 5] \text{ N/s}$.
- These findings indicate that haptic force sensitivity improved as the force-rate is increased.